port numbers

we run multiple programs on a machine

- IP addresses identifying machine — not enough

port numbers

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- IP addresses identifying machine — not enough

so, add 16-bit *port numbers*

- think: multiple PO boxes at address

port numbers

we run multiple programs on a machine

- IP addresses identifying machine — not enough

so, add 16-bit *port numbers*

- think: multiple PO boxes at address

0–49151: typically assigned for particular services

- 80 = http, 443 = https, 22 = ssh, ...
- usually <1024 reserved for sysadmin-only-use

49152–65535: allocated on demand

- default "return address" for client connecting to server

port number independence

typically "5-tuple" identifies socket:
(protocol=TCP or UDP, source IP, dest IP, source port, dest port)

means can have:

- two sockets with same source IP+source port, but different destinations
- two sockets with same dest IP+des port, but different sources
- two sockets with different protocols, but everything else same

special cases:

- dest IP/port can be 'wildcard' (all zeroes usually)
- used for server that can be connected to
- used for unconnected UDP sockets

UDP format

(IPv4/IPv6 header)

source port	destination port
length	checksum

(application data)

BSD sockets

BSD = Berkely Standard Distribution (of Unix)

interface designed for stream connections

current C version standardized by POSIX

pretty much default interface programming with TCP connections

...across all platforms

sockets as generic interface

sockets designed to be protocol-agnostic

support all sorts of protocols with addresses

support streams and datagrams

unconnected UDP sockets (Python)

```
import socket
# ipv4
sock = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
sock = socket.socket(socket.AF_INET, socket.SOCK_DGRAM,
                     socket.IPPROTO_UDP)
# ipv6
sock = socket.socket(socket.AF_INET6, socket.SOCK_DGRAM)
sock = socket.socket(socket.AF_INET6, socket.SOCK_DGRAM,
                     socket.IPPROTO_UDP)

sock.bind((local_host, local_port))
# local_host == "0.0.0.0" (v4) or "::" (v6) for "all possible

data1, (host1, port1) = sock.recvfrom(MAX_SIZE)
sock.sendto(data, (host2, port2))
```

unconnected UDP sockets (Python, generic)

```
import socket
possible_local_addrs = socket.getaddrinfo(
        local_host, local_port,
        family=socket.AF_UNSPEC,
        type=socket.SOCK_DGRAM,
        flags=socket.AI_PASSIVE)

for af, socktype, proto, canonname, sockaddr in possible_local_addrs:
    sock = socket.socket(af, socktype, proto)
    sock.bind(sockaddr)
    # then use sock for recvfrom/sendto somehow
```

unconnected UDP sockets (Python, generic)

```
import socket
possible_remote_addrs = socket.getaddrinfo(
        remote_host, remote_port,
        family=socket.AF_UNSPEC,
        type=socket.SOCK_DGRAM,
        flags=0)

for af, socktype, proto, canonname, sockaddr in possible_remote_addrs:
    sock = socket.socket(af, socktype, proto)
    try:
        # hope local_host is acceptable for 'af'?
        # maybe choose local_host based on 'af' value?
        sock.bind((local_host, local_addr))
        sock.sendto(data, ...)
        ...
    except ...:
        ...
        # hope another address works?
        continue
```

aside: getaddrinfo

lookup host+port by name

can handle both IPv4 and IPv6

AI_PASSIVE = for bind()

returns list of addresses

- sometimes: just choose one address (send message to server)
- sometimes: want to make socket for each address (run server)

unconnected UDP sockets (C, IPv4)

```
int fd = socket(AF_INET, SOCK_DGRAM, 0 /* or IPPROTO_UDP */);
/* for 10.1.2.3 port 12345
   there are utility functions to help with this, so you
   shouldn't actually construct local_addr manually like this */
struct sockaddr_in local_addr;
memset(&local_addr, 0, sizeof(local_addr);
local_addr.sin_family = AF_INET;
local_addr.sin_port = htons(12345);
local_addr.sin_addr.s_addr = htonl((10<<24)|(1<<16)|(2<<8)|3),
bind(fd, &local_addr, sizeof(local_addr));
struct sockaddr_in remote1, remote2;
recvfrom(fd, buffer1, buffer1_size, 0, &remote1, sizeof(remote1));
recvfrom(fd, buffer2, buffer2_size, 0, &remote2, sizeof(remote2));
struct sockaddr_in remote3 = ...;
sendto(fd, buffer3, buffer3_size, 0, &remote3, sizeof(remote3));
```

connected UDP sockets (Python)

```
import socket
# ipv4
sock = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
# ipv6
sock = socket.socket(socket.AF_INET6, socket.SOCK_DGRAM)

sock.bind((local_host, local_port))
sock.connect((remote_host, remote_port))
sock.send(b"one datagram")
sock.send(b"another datagram")
remote_data1 = sock.recv(MAX_SIZE)
remote_data2 = sock.recv(MAX_SIZE)
```

choosing v4/v6?

nicer to write applications that handle both

...without explicitly doing this differently

later socket API for that is getaddrinfo()

also exists in C, but lots of extra stuff to handle types, etc.

netstat

```
$ netstat --udp -na
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
udp        0      0 128.143.71.27:60001     0.0.0.0:*
udp        0      0 128.143.71.27:60002     0.0.0.0:*
udp        0      0 128.143.71.27:60003     0.0.0.0:*
...
udp        0      0 128.143.71.27:68        128.143.67.64:67        ESTABLISHED
...
udp        0      0 127.0.0.1:123           0.0.0.0:*
udp        0      0 0.0.0.0:38269           0.0.0.0:*
udp        0      0 0.0.0.0:55407           0.0.0.0:*
udp        0      0 0.0.0.0:55786           0.0.0.0:*
...
udp6       0      0 :::45631                :::*
udp6       0      0 :::111                  :::*
udp6       0      0 fe80::1a66:daff:fe2:123 :::*
...
```

connection setup: client, TCP+IPv4

```
sock = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
# to select local host/port: socket.bind(('128.143.71.27', 54444))
# if no local host/port selected, OS assigns
sock.connect(('128.143.67.8', 443))
...
sock.send(...)
some_bytes = sock.recv(max_size)
```

connection setup: client, TCP+IPv6

```
sock = socket.socket(socket.AF_INET6, socket.SOCK_STREAM)
# to select local host/port: socket.bind(('3fff:1234::1', 54444))
# if no local host/port selected, OS assigns
sock.connect(('2607:f8b0:4004:c06::67', 443))
...
sock.send(...)
some_bytes = sock.recv(max_size)
```

connection setup: generic TCP, addrinfo

```
import socket
possible_addrs = socket.getaddrinfo(
        'www.cs.virginia.edu', '443',
        socket.AF_UNSPEC, socket.SOCK_STREAM)

for af, socktype, proto, canonname, sockaddr in possible_addrs:
    try:
        sock = socket.socket(af, socktype, proto)
    except OSError:
        sock = None
        continue
    try:
        sock.connect(sa)
    except OSError:
        sock.close()
        sock = None
        continue

if sock == None:
    # not successful
```

size mismatch? (1)

one end

other end

(get fds/socks:)

```
sock.send(b"12345")
```



```
b"123" == sock.recv(3)
b"45" == sock.recv(2)
```

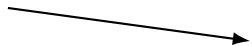
size mismatch? (2)

one end

other end

(get fds/socks:)

```
sock.send(b"12")
sock.send(b"345")
```



```
b"12345" == sock.recv(5)
```

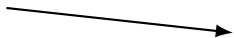
partial reads?

one end

other end

(get fds/socks:)

```
sock.send(b"X" * 6000)
```



sometimes `b"X" * 6000 == sock.recv(6000)`
or sometimes

```
b"X" * 1490 == sock.recv(6000)
b"X" * 2980 == sock.recv(4510)
b"X" * 1530 == sock.recv(1530)
```

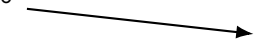
partial writes

one end

other end

(get fds/socks:)

```
sock.setblocking(False)
sock.send(b"X" * 6000) == 1480
```



```
b"X" * 1480 == sock.recv(6000)
(+ no more data)
```

partial reads/writes

read/recv can have read less than requested

- read of zero bytes = end of file

send/write can write less than requested...
but only on error or if non-blocking mode requested

incomplete writes

`send` might write less than requested

- error after writing some data
- if blocking disabled

`recv` might read less than requested

- error after reading some data
- not enough data got there in time

reading and writing at once

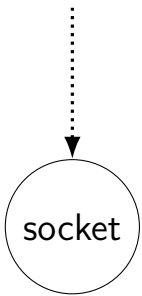
threads: one reading thread, one writing thread *OR*

event-loop: use *non-blocking I/O* and select()/poll()/etc. functions

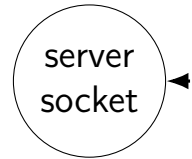
- select/poll/etc. tell you when one of many operations possible
- ...including across many sockets

sockets and server sockets (C)

```
client:
fd = socket(…)
```



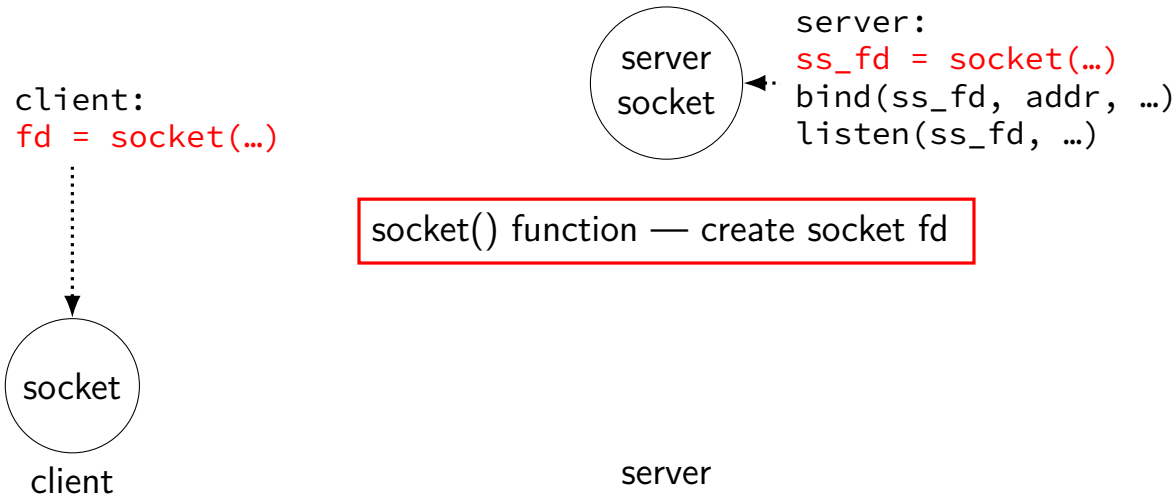
client



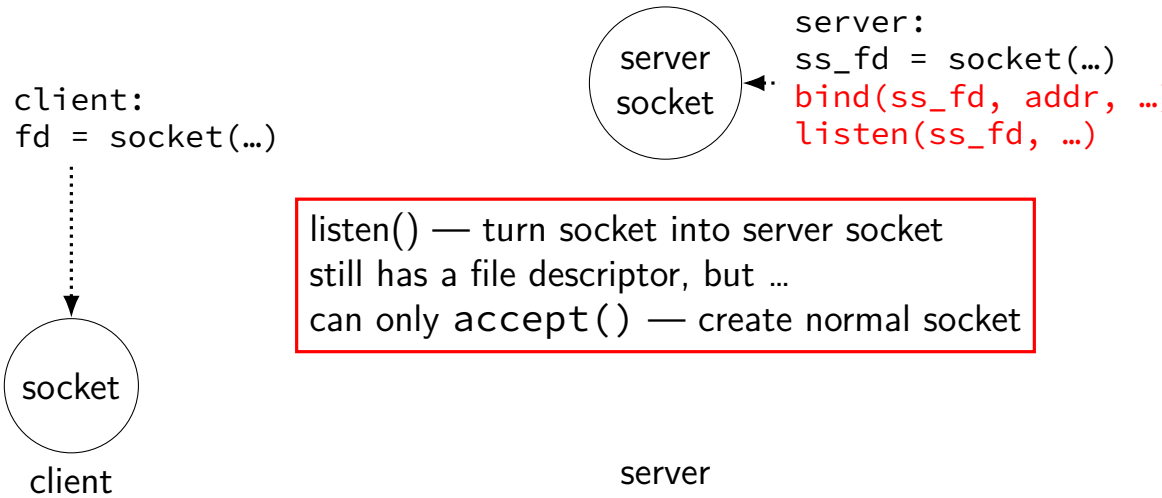

```
server:
ss_fd = socket(…)
bind(ss_fd, addr, …)
listen(ss_fd, …)
```

server

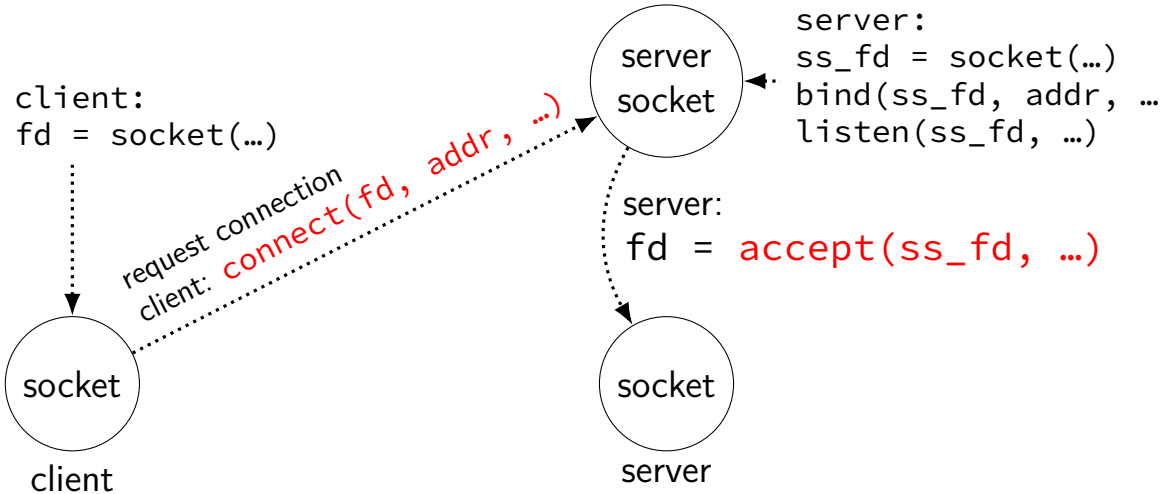
sockets and server sockets (C)



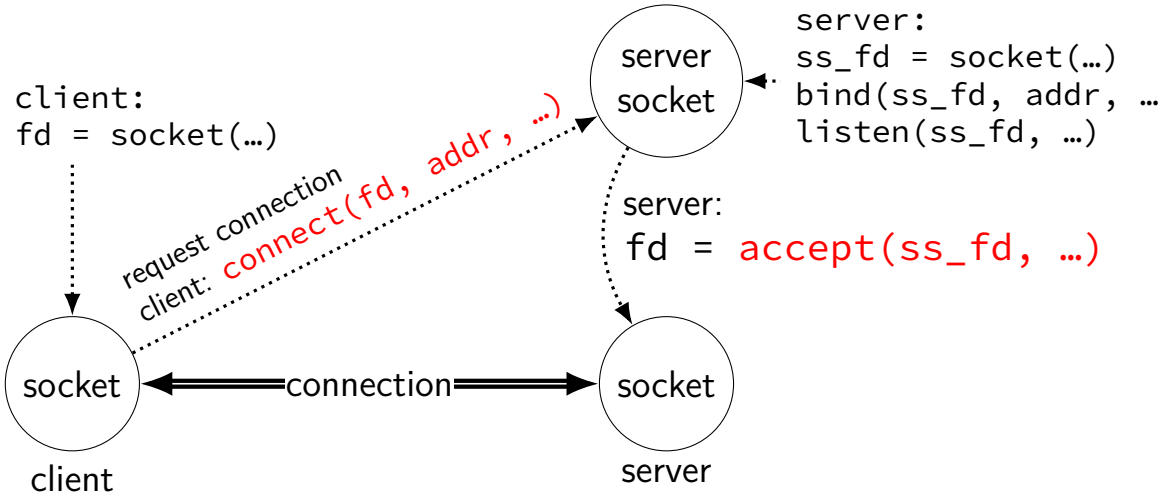
sockets and server sockets (C)



sockets and server sockets (C)

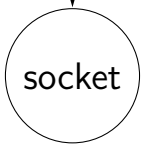


sockets and server sockets (C)

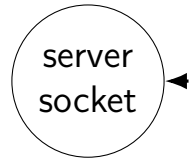


sockets and server sockets (Python)

```
client:
sock = socket.socket(…)
```



client

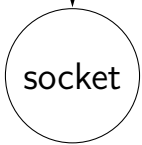



```
server:
ssock = \
 socket.socket(…)
ssock.bind(…)
ssock.listen()
```

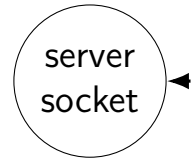
server

sockets and server sockets (Python)

```
client:
sock = socket.socket(…)
```



client

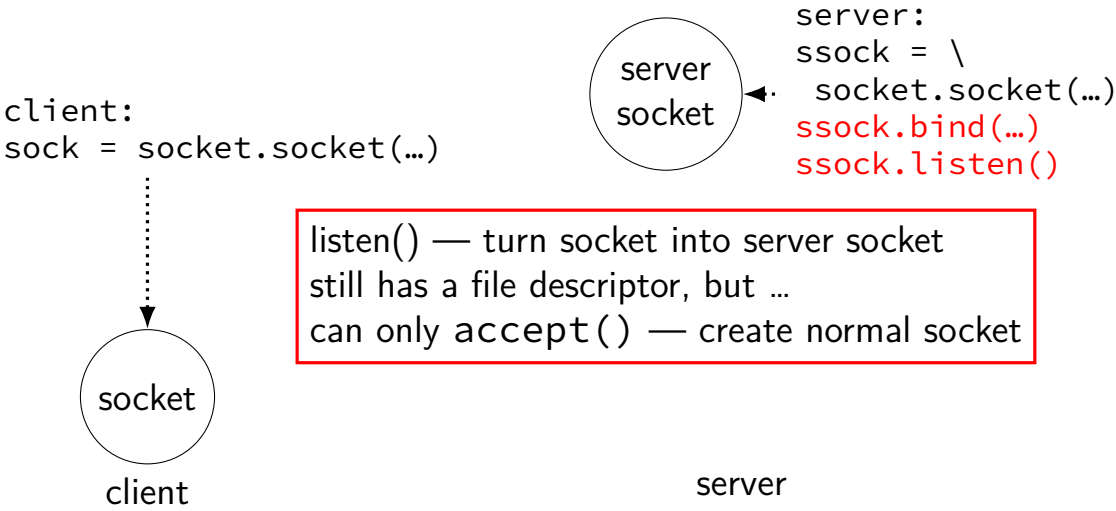



```
server:
ssock = \
 socket.socket(…)
ssock.bind(…)
ssock.listen()
```

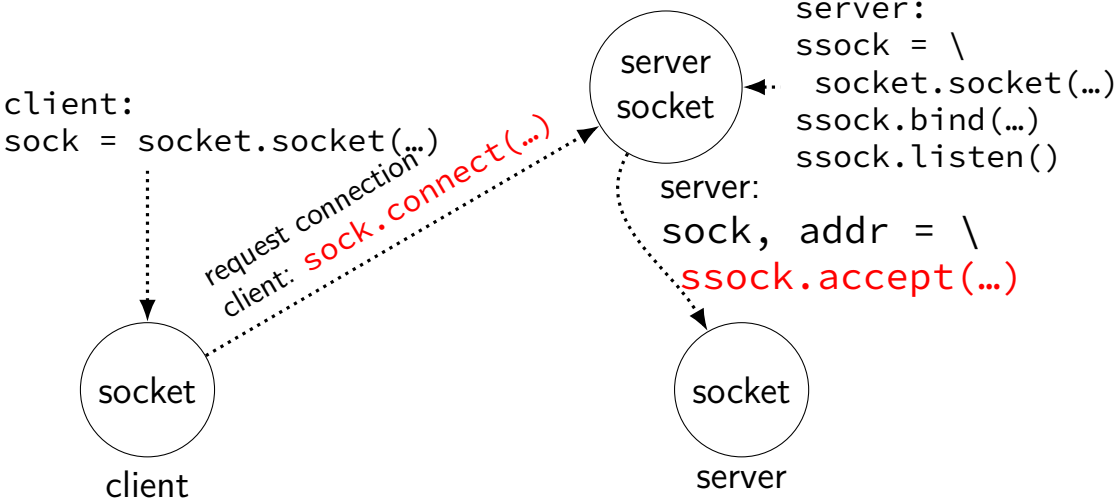
socket() function — create socket fd

server

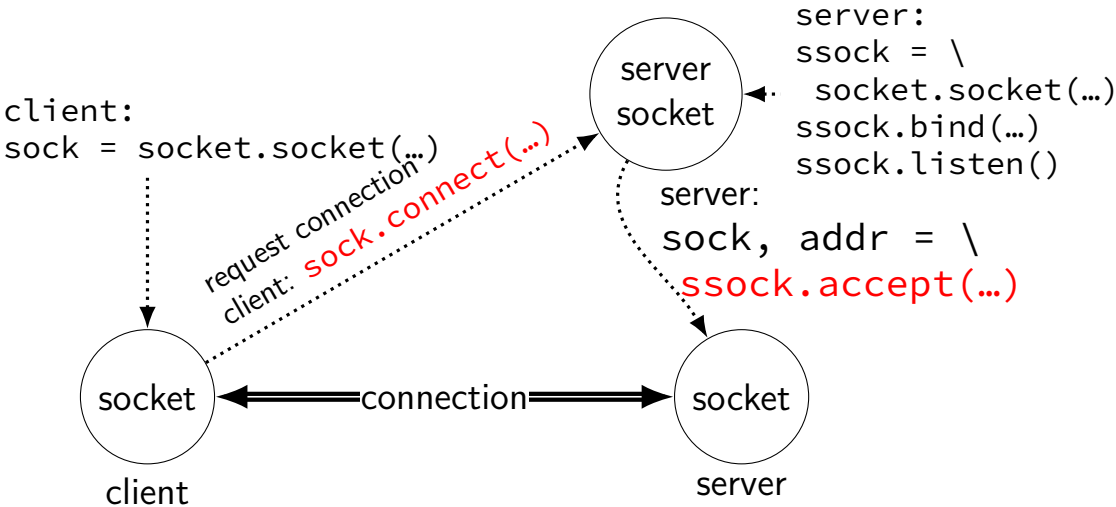
sockets and server sockets (Python)



sockets and server sockets (Python)



sockets and server sockets (Python)



connection setup: server, IPv4

```
import socket
ssock = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
# 0.0.0.0 == "any IP address"
ssock.bind(('0.0.0.0', 12345))
ssock.listen()
sock, remote_addr = ssock.accept()
Use(sock)
```

connection setup: server, IPv4

```
import socket
ssock = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
# 0.0.0.0 == "any IP address"
ssock.bind(('0.0.0.0', 12345))
ssock.listen()
sock, remote_addr = ssock.accept()
Use(sock)
```

connection setup: server, generic

```
import select
import socket
all_addrs = socket.getaddrinfo(
        None, '12345',
        socket.AF_UNSPEC, socket.SOCK_STREAM)
ssocks = []
for af, socktype, proto, canonname, sockaddr in possible_addrs:
    ssock = socket.socket(af, socktype, proto)
    try:
        ssock.bind(sockaddr)
        ssock.listen()
        ssocks.append(ssock)
    except OSError:
        pass

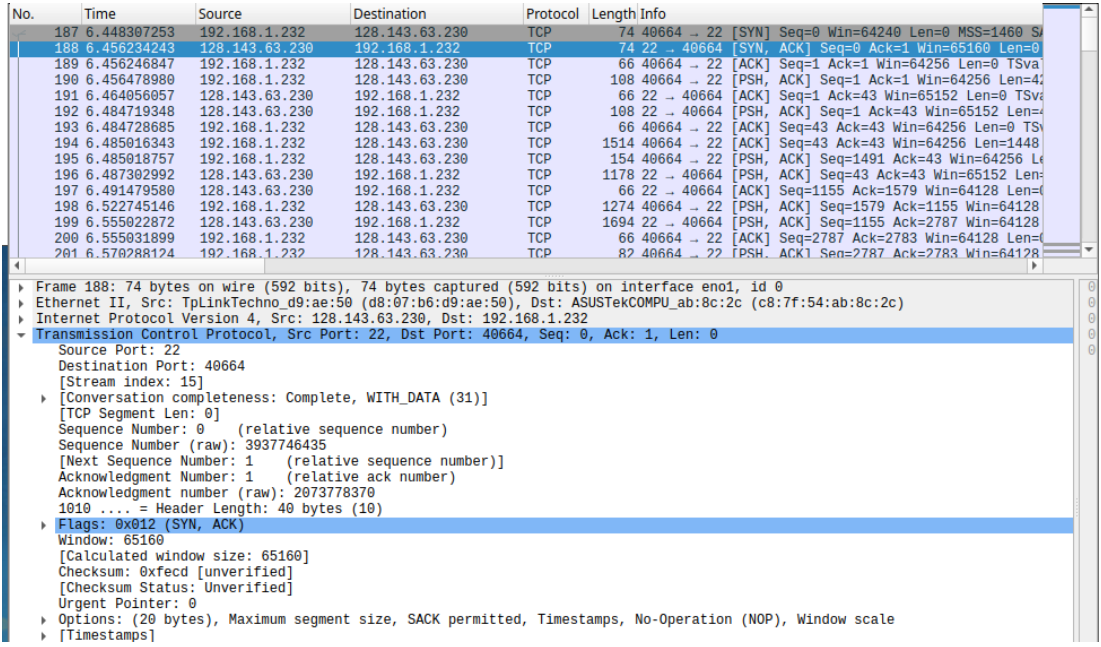
which_ready, _, _ = select.select(ssocks, [], [])
for item in which_ready:
    sock, remote_addr = which_ready.accept()
    Use(sock)
```

three-way TCP handshake

from →to	flags	seq	ack	data
client →server	SYN	X	—	—
server→client	SYN+ACK	Y	$X+1$	—
client →server	ACK	$X+1$	$Y+1$	—
client →server	ACK	$X+1$	$Y+1$	initial data

X, Y selected at random

random selection to prevent attacker from 'spoofing' connection by sending packets with forged source addresses + guessing what to ACK


No. Time Source Destination Protocol Length Info
187 6.448307253 192.168.1.232 128.143.63.230 TCP 74 40664 → 22 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 S
188 6.456234243 128.143.63.230 192.168.1.232 TCP 74 22 → 40664 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0
189 6.456246847 192.168.1.232 128.143.63.230 TCP 66 40664 → 22 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSva
190 6.456478980 192.168.1.232 128.143.63.230 TCP 108 40664 → 22 [PSH, ACK] Seq=1 Ack=1 Win=64256 Len=4
191 6.464056057 128.143.63.230 192.168.1.232 TCP 66 22 → 40664 [ACK] Seq=1 Ack=43 Win=65152 Len=0 TSv
192 6.484719348 128.143.63.230 192.168.1.232 TCP 108 22 → 40664 [PSH, ACK] Seq=1 Ack=43 Win=65152 Len=
193 6.484728685 192.168.1.232 128.143.63.230 TCP 66 40664 → 22 [ACK] Seq=43 Ack=43 Win=64256 Len=0 TS
194 6.485016343 192.168.1.232 128.143.63.230 TCP 1514 40664 → 22 [ACK] Seq=43 Ack=43 Win=64256 Len=1448
195 6.485018757 192.168.1.232 128.143.63.230 TCP 154 40664 → 22 [PSH, ACK] Seq=1491 Ack=43 Win=64256 Le
196 6.487302992 128.143.63.230 192.168.1.232 TCP 1178 22 → 40664 [PSH, ACK] Seq=43 Ack=43 Win=65152 Len
197 6.491479580 128.143.63.230 192.168.1.232 TCP 66 22 → 40664 [ACK] Seq=1155 Ack=1579 Win=64128 Len=
198 6.522745146 192.168.1.232 128.143.63.230 TCP 1274 40664 → 22 [PSH, ACK] Seq=1579 Ack=1155 Win=64128
199 6.555022872 128.143.63.230 192.168.1.232 TCP 1694 22 → 40664 [PSH, ACK] Seq=1155 Ack=2787 Win=64128
200 6.555031899 192.168.1.232 128.143.63.230 TCP 66 40664 → 22 [ACK] Seq=2787 Ack=2783 Win=64128 Len=
201 6.570288124 192.168.1.232 128.143.63.230 TCP 82 40664 → 22 [PSH, ACK] Seq=2787 Ack=2783 Win=64128
Frame 188: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface eno1, id 0
Ethernet II, Src: TpLinkTechno_d9:ae:50 (d8:07:b6:d9:ae:50), Dst: ASUSTekCOMPU_ab:8c:2c (c8:7f:54:ab:8c:2c)
Internet Protocol Version 4, Src: 128.143.63.230, Dst: 192.168.1.232
Transmission Control Protocol, Src Port: 22, Dst Port: 40664, Seq: 0, Ack: 1, Len: 0
Source Port: 22
Destination Port: 40664
[Stream index: 15]
[Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 0]
Sequence Number: 0 (relative sequence number)
Sequence Number (raw): 3937746435
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 2073778370
1010 .... = Header Length: 40 bytes (10)
Flags: 0x012 (SYN, ACK)
Window: 65160
[Calculated window size: 65160]
Checksum: 0xfecd [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
Options: (20 bytes), Maximum segment size, SACK permitted, Timestamps, No-Operation (NOP), Window scale
[Timestamps]

so many messages?

TCP waits full round-trip before sending data; can do better

example: QUIC (figure from RFC 9000, TCP+TLS replacement)

```
Client                                                  Server

Initial[0]: CRYPTO[CH]
0-RTT[0]: STREAM[0, "..."] ->

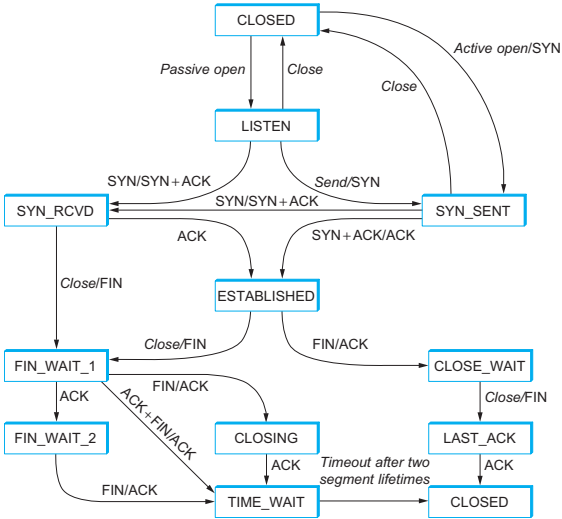
                                  Initial[0]: CRYPTO[SH] ACK[0]
                                   Handshake[0] CRYPTO[EE, FIN]
                          <- 1-RTT[0]: STREAM[1, "..."] ACK[0]

Initial[1]: ACK[0]
Handshake[0]: CRYPTO[FIN], ACK[0]
1-RTT[1]: STREAM[0, "..."] ACK[0] ->

                                               Handshake[1]: ACK[0]
            <- 1-RTT[1]: HANDSHAKE_DONE, STREAM[3, "..."], ACK[1]
```

Figure 6: Example 0-RTT Handshake

TCP state machine



notation: event/action

connection close

FIN (I want to close) + ACK (confirm close)

ACK side needs to wait just in case FIN resent

- keep port number from being reused
- TIME_WAIT state

RST

RST (reset) flag often used to respond to unexpected packets

example: data packet for never-established connection

connections on my desktop

```
cr4bd@reiss-t3620
: /zf14/cr4bd ; netstat --inet --inet6 --numeric
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 128.143.67.91:49202     128.143.63.34:22        ESTABLISHE
tcp        0      0 128.143.67.91:803       128.143.67.236:2049     ESTABLISHE
tcp        0      0 128.143.67.91:50292     128.143.67.226:22       TIME_WAIT
tcp        0      0 128.143.67.91:54722     128.143.67.236:2049     TIME_WAIT
tcp        0      0 128.143.67.91:52002     128.143.67.236:111      TIME_WAIT
tcp        0      0 128.143.67.91:732       128.143.67.236:63439    TIME_WAIT
tcp        0      0 128.143.67.91:40664     128.143.67.236:2049     TIME_WAIT
tcp        0      0 128.143.67.91:54098     128.143.67.236:111      TIME_WAIT
tcp        0      0 128.143.67.91:49302     128.143.67.236:63439    TIME_WAIT
tcp        0      0 128.143.67.91:50236     128.143.67.236:111      TIME_WAIT
tcp        0      0 128.143.67.91:22        172.27.98.20:49566      ESTABLISHE
tcp        0      0 128.143.67.91:51000     128.143.67.236:111      TIME_WAIT
tcp        0      0 127.0.0.1:50438         127.0.0.1:631           ESTABLISHE
tcp        0      0 127.0.0.1:631           127.0.0.1:50438         ESTABLISHE
```

socket options

'socket options' for extra settings

Python: `sock.setsockopt(level, key, value)`

e.g. `sock.setsockopt(socket.SOL_SOCKET, socket.SO_REUSEADDR, 1)`

C: `setsockopt(sock, level, key, value)`

e.g. `setsockopt(sock, SOL_SOCKET, SO_REUSEADDR, 1)`

common level values

- `SOL_SOCKET` — OS-level
- `IPPROTO_IP`
- `IPPROTO_IPV6`
- `IPPROTO_TCP`

common SOL_SOCKET options

SO_REUSEADDR — allow bind()ing even if another socket for this port active

- common problem: server can't bind() because old session in TIME_WAIT

SO_RCVBUF — buffer size used by OS for receiving bytes

SO_SNDBUF — buffer size used by OS for sending bytes

SO_LINGER — make close() wait until messages acknowleged or timeout

(full list on Linux: `man 7 socket`)

efficiency problem

```
sock = ...
sock.send(b'About to send file with 16Kbytes')
sock.send(b'File is HTML; updated 16 November')
sock.send(file_data)
```

Naive approach:

- one packet for ‘about to send file with...’
- one packet for ‘file is HTML; ...’
- some number of packets for `file_data`

sending small TCP packets, so lots of overhead

Nagle's algorithm

(RFC 896)

delay sending data when smaller than full TCP segment

hope: by waiting we'll get a full TCP segment

typical wait: 200 ms?

TCP_NODELAY socket option can disable (at least on Linux)

broadcast/multicast

IPv4: can broadcast to every machine on local network

- set the SO_BROADCAST socket option to 1
- send to special address 255.255.255.255
- received at that port number on all machines

multicast (send-to-many) groups

- request to receive: IP_ADD_MEMBERSHIP (v4), IPV6_ADD_MEMBERSHIP (v6)
- each 'multicast group' has IP address
- 224.0.0.0/24 and ff02::/16 = local network only
- local network version usually implemented by broadcasting + filtering by IP
- (non-local-network multicast is more complex...)

service discovery

example: find printers on local network automatically

typical protocol mDNS ('multicast DNS')

uses IP addressses 224.0.0.251 / ff02::fb + port 5353

all machines on local network receive from all other machines on local network

other address families

AF_UNIX (= AF_LOCAL in C, but not Python)

- sockets can be represented in files on disk
- support sending file descriptors usually

(not sure how portable?) AF_RAW

- usually how wireshark captures packets on Linux
- also supports writing packets

(Linux only) AF_NETLINK

- communicate with kernel networking code

AF_AX25, AF_APPLETALK, AF_DECNET, ...

- other interesting protocols

backup slides